

A Harmonic Dualism Model for *Diatonic Sets* and Temperaments from Lewin's *Formal Theory*

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Lewin (1982) generalised the behavior of tonal functions out of consonant harmonies into a formal model, that codified the relations between triads and pitches originally identified by Rameau (1726) and Riemann (1896).

Lewin initiated the transformational view of harmony and analysed the specific relations allowing harmonic dualism within a group of transformations called the *serial type-group*. However, two limitations are significant for analysing harmonic relations in a dualistic fashion:

- 1) All unordered pitch dispositions derived from the formal structure – the *Riemann Systems* (R.S.) – can be “directed above” or “below”, but not both (Lewin 1982, 57–58);
- 2) Intonation/temperament matters when transformations occur within the *shift group* of transformations, which encompasses all functional relations (Lewin 1982, 50–51).

The ordered pitch disposition of an R.S.—the *diatonic set*—follows the characteristic rearrangement of functional triads' pitches in succession (i.e.: $\hat{1}$ from Tonic, $\hat{2}$ from Dominant, $\hat{3}$ from Tonic, $\hat{4}$ from Subdominant, $\hat{5}$ from Dominant, $\hat{6}$ from Subdominant, and $\hat{7}$ from Dominant). The pitches' “functions” depend closely on the triads that generated them, and their intonation depends on the interval they form with their tonic pitch (Lewin 1982; Harrison 2010).

Lewin's triads are formed by applying a **m** (mediant) and a **d** (dominant) interval to a **T** (tonic) pitch. These intervals must be distinct and non-zero. The interval **d** - **m** forms the **m'** interval. In the change of disposition or ordering of these intervals lies the change of triads' mode and harmonic dualism, and the R.S. is derived from the disposition of pitches in the ordered Subdominant-Tonic-Dominant triads.

My proposal modifies the model to create a system of pitch disposition that follows Lewin's theory while forming triads from pitches. To achieve this, the pitches must utilise Lewin's interval $\mathbf{m}'$ and the difference $\mathbf{m} - \mathbf{m}' = \partial$ (the "semitone"):

$$d = m + m' = 2m' + \partial$$

$$\text{major triad} = [T, d, m] = [T, d, m' + \partial]$$

$$\text{minor triad} = [T, d, (d - m)] = [T, d, m']$$

All Lewin's transformations remain valid, but the change of mode can also be executed within the set via the semitone interval ∂ .

The conditions for this model are summarised as:

$$0 < \partial < m' < d < \text{one octave}.$$

This model allows for the application of Lewin's model to all possible pitch combinations within a pitch-set—in any intonation that respects the conditions—and enables the R.S. to be conceptually "directed" both "above" and "below".

The following examples of outcomes underscore how the symmetries of harmonic dualism are reflected in various diatonic set features:

12 EDO (Equal Division of Octave): Highlights how **R** and **L** neo-Riemannian transformations represent the same formal relation as tonality's **P** when the R.S. forms triads in second inversion;

24 EDO: Highlights different features of enharmonic equivalents;

31 EDO: A set with few enharmonic equivalents, it reveals a compelling R.S., very close to tonality, that lies exactly between the equal division of the fifth into two "neutral" thirds and the "standard" tonal R.S. (Gann 2019).

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